

ReAL-Health: Reliability and Assurance for Language Technology in Health

A Workshop Proposal for NAACL 2027

Making clinical language technology reliable, auditable, private, and safe enough to deploy.

Short name ReAL-Health. **URL** <https://real-health-workshop.github.io/> **Duration** one day.

Attendance 100–150; 30–40 posters.

Venue NAACL 2027 (first), EMNLP 2027 (second); no incompatible venues. **Prior editions** none — first edition.

1. Topic, Motivation, and Relevance

Clinical language technology is already deployed at scale: ambient scribes draft notes, LLMs triage patient messages, summarize discharge records, and increasingly act as agents over the electronic health record (EHR) [1, 2, 3]. But what makes a model accurate on a clinical QA benchmark is not what makes it safe at the bedside. Deployment demands properties our field treats as secondary: calibrated abstention, faithful grounding in the patient record, robustness to the messy multilingual language clinicians and patients actually produce, guarantees over protected health information (PHI), and evidence a regulator can audit.

ReAL-Health asks one question: *how do we build clinical language technology — increasingly agentic — reliable enough to deploy, and how do we evidence it?* Three clusters:

- **Reliability and evaluation of clinical NLP:** uncertainty, calibration, and selective prediction in clinical generation [4]; hallucination and faithfulness in EHR-grounded summarization [5, 19]; clinician-in-the-loop and rubric-based evaluation beyond accuracy; negative results and failed deployments.
- **Safety and assurance for clinical LLM agents:** long-horizon evaluation of agents acting over the EHR; faithfulness of tool calls against FHIR resources; prompt injection via patient-supplied text and retrieved records [10]; red-teaming methodology; escalation and human-handoff design; audit logs as assurance artifacts; assurance cases translating the EU AI Act [11], FDA SaMD guidance [12], and NIST AI RMF [13] into NLP evaluation protocols.
- **Privacy, fairness, and governance for clinical text:** de-identification and re-identification risk [6]; PHI memorization and leakage; federated training across hospital corpora [7]; HIPAA/GDPR-aligned unlearning [8]; disparate performance across patient populations [9].

Multilingual and low-resource clinical NLP cuts across all three: reliability established on English discharge summaries does not transfer to the languages in which most of the world receives care. We solicit work on non-English corpora, code-switched clinician–patient interaction, and interpretation.

Audience and timing. Clinical NLP, reliability and safety researchers; informaticians deploying inside health systems; researchers in audit and regulation. Deployment has outpaced assurance: documented failures — hallucinated dosages [5], bias in risk scoring [9], injection through patient-authored text [10] — arrived alongside institutional rollout, and regulation now classifies clinical decision support as high-risk [11, 12]. NLP owns the evaluation methodology this demands but publishes it apart from the clinical and audit communities that must use it. We therefore reserve time for deployment experience reports and regulatory translation, excluded from paper-track-only venues.

2. Differentiation from Existing Venues

ClinicalNLP and **BioNLP** center tasks, models, and resources for clinical text; **LOUHI** centers health text analytics broadly; **TrustNLP** treats reliability, privacy, and fairness as general NLP properties. ReAL-Health occupies the gap: *assurance evidence* for clinical language systems already leaving the lab, operationalized against HIPAA, GDPR, FDA SaMD, and the EU AI Act. Topics we solicit that those venues do not routinely host: (i) assurance cases and audit artifacts for NLP systems; (ii) threat modeling and red-teaming for EHR-integrated agents; (iii) unlearning and right-to-be-forgotten for clinical LMs; (iv) deployment post-mortems and negative results from health systems; (v) regulator-facing evaluation protocol design. Agentic health workshops have begun to appear, but are anchored in imaging and multimodal data (AIM at NeurIPS 2026; Agentic AI for Medicine at MICCAI 2026) or in general autonomy; none sits in a CL venue or centers the language-native failure modes of EHR agents.

3. Format, Schedule, and Discussion

One day, roughly 7.5 hours, with more than half the time in discussion, posters, or panel rather than podium talks. **Schedule.** 0:00 opening, code of conduct and anti-harassment policy (10 min); 0:10 *Keynote I*, health-system deployment (40); 0:50 *Contributed I*, reliability, grounding, evaluation (55); 1:45 *Posters I*, protected, no parallel content (60); 2:45 *Keynote II*, methods (40); 3:25 *Panel* across research, hospitals, regulators (55); 4:20 *Contributed II*, agent safety, privacy, fairness (55); 5:15 shared-task spotlights (30); 5:45 *Posters II* (60); 6:45 facilitated open-problems discussion drafting the community memo (35); 7:20 awards and closing (10).

Hybrid and virtual-only contingency. The workshop can run fully virtually without loss. Every accepted paper supplies a pre-recorded video released beforehand, so discussion is not gated on live attendance. Talks and the panel are live-streamed with a *dedicated remote moderator* whose only role is relaying chat questions into the room, so remote

attendees are not competing with in-room microphones. Posters run on a virtual platform with author slots duplicated across Americas- and Europe/Asia-friendly time bands, so no region is confined to asynchronous participation. Panel and discussion questions are collected in advance via a public form, equalizing remote and in-person input. If the venue is virtual-only, the program splits into two half-days in complementary time zones; the shared task is unaffected.

Shared task: safety of patient-facing clinical LLMs. Built on the HealthBench rubric [14]. Track (a): a guardrail predicting rubric-graded safety of a response. Track (b): red-teaming — prompts eliciting unsafe clinical responses from a held-out reference model. Multilingual and multi-turn agentic subtracks extend a prompt subset beyond English and beyond single responses; we estimate 25–40 teams. Released publicly: top guardrail models, a community red-team set, and the evaluation harness. Announced with the first call (26 Oct 2026); baselines December; evaluation 8 Feb–1 Mar 2027; descriptions 8 Mar; notification 26 Mar.

Ethics and risk. Shared-task data are synthetic or publicly licensed; no real PHI is used or accepted. The red-team track releases prompts but withholds completions constituting directly actionable clinical harm, under a posted responsible-release policy. Dual-use handling follows the ACL Code of Ethics.

4. Submissions, Review, and Requirements

Call. Long (8 pages) and short (4 pages) papers, unlimited references, ACL style, anonymized: novel research, resources, deployment experience, negative results, position papers. **Archival** proceedings in the ACL Anthology, plus a **non-archival** option (Findings, work under review elsewhere) for presentation only — important for hospital and industry contributors publishing in clinical venues.

Review platform. We will use **OpenReview as our own reviewing platform and will not accept ARR-committed submissions**. For a first edition whose audience includes clinicians, informaticians, and regulatory researchers largely outside the ARR ecosystem, a single direct route is simpler and lets us apply a rubric tailored to assurance claims. Double-blind, three PC members per paper, scoring claim–evidence alignment separately from novelty.

Timeline (suggested NAACL schedule): calls 26 Oct and 23 Nov 2026; submissions 5 Feb 2027; notification 26 Mar; camera-ready 16 Apr; proceedings 7 May; videos 14 May — we commit to both hard deadlines. The CFP will be ready one week after notification.

Volume, PC capacity, requirements. We estimate 50–70 submissions. The 38 confirmed PC members cover about 38 papers at three reviews each with nobody exceeding three, so we are expanding the PC to 70+ before the first call, recruiting in clinical NLP and at European, Asia-Pacific, and Global South institutions. Needs: one room seating 100–150 with standard projection and microphones; 30–40 poster boards; live-stream and recording as provided. Reliable wifi is the only non-standard need.

Conflicts of interest. Organizers do not submit or present, review submissions from their own institutions, or handle work by current or recent students. The shared-task lead is recused from systems with co-authors at her institution; documented in OpenReview.

5. Invited Speakers and Funding

Already agreed (9). *Keynotes:* Xiaofeng Wang (Cleveland Clinic); Anant Madabhushi (Georgia Tech and Emory). *Invited talks:* Robert Davis (UTHSC); Vipin Chaudhary (CWRU); Ritambhara Singh (Brown); Satish E. Viswanath (Emory); Farshid Alambeigi (UT Austin); Shuo Li (CWRU); Heidi Hanson (ORNL). **Tentative:** invitations to clinical-NLP and regulatory-affairs speakers are in progress. **Funding:** the conference’s single free registration goes to one keynote; all others are locally based or self-funded from institutional and grant funds, confirmed individually, so no further cost falls to the conference. Sponsorship, if secured, funds shared-task prizes and student subsidies.

6. Diversity and Inclusion

Academic diversity. The subject matter is itself a contribution: reliability evidence for clinical NLP is overwhelmingly established on English records from a few well-resourced US and European health systems and does not demonstrably transfer. The shared task carries a multilingual subtrack, and all artifacts are permissively licensed so groups without proprietary data can build on them.

Diversifying representation. We acknowledge a real gap: six of seven organizers are men and all are at US institutions. We are recruiting an additional co-organizer who is a woman based outside the US (the call permits eight) and weighting PC expansion accordingly. The confirmed slate includes two women (Singh, Hanson) and spans academic, clinical, and national-laboratory settings, and none has keynoted a recent ACL-family conference. Remaining invitations prioritize women, non-US researchers, and clinicians.

Diversifying participation. The call invites clinicians, deployment practitioners, and researchers from low-resource settings, and puts experience reports and negative results in scope — lowering the barrier for contributors without large compute budgets. A mentoring program pairs first-time submitters from under-represented groups with PC members for pre-submission feedback. We coordinate outreach with WiNLP and affinity groups and direct any sponsorship raised to registration subsidies for students and Global South participants, whom conference D&I chairs do not cover.

Organizer Information

Organizing Committee

Md Tauhidul Islam	Stanford University, USA	tauhid@stanford.edu
Avanti Bhandarkar	Mayo Clinic, USA	bhandarkar.avanti@mayo.edu
Sumon Biswas	Case Western Reserve University, USA	sumon@case.edu
Amit Kumar Sikder	Iowa State University, USA	aksikder@iastate.edu
Shahnewaz Karim Sakib	Univ. of Tennessee at Chattanooga, USA	shahnewazkarim-sakib@utc.edu
Anindya Bijoy Das	University of Akron, USA	adas@uakron.edu
Shibbir Ahmed	Texas State University, USA	shibbir@txstate.edu

Biographies and Workshop Experience

Md Tauhidul Islam (Stanford) is Assistant Professor of Radiation Oncology, Stanford Medicine, working on foundation models for multimodal biomedical data and on clinical translation, including in-context learning for medical report generation [15] — directly a question of grounding and faithfulness in clinical generation. *Role:* program co-chair.

Avanti Bhandarkar (Mayo Clinic) is a Research Fellow with 8+ years designing computational models for clinical text understanding, predictive healthcare modeling, and secure AI systems [16]. PhD, University of Florida, in generative AI and computational behavioral modeling. *Role:* shared-task lead; clinical-text and health-system deployment perspective.

Sumon Biswas (Case Western Reserve) is Assistant Professor of Computer and Data Sciences working on responsible AI engineering, verification, and algorithmic fairness [17], with NSF and DARPA support. PCs at ICSE, FSE, ASE; Accessibility Chair, ACM SPLASH 2020–21; Board of Distinguished Reviewers, ACM TOSEM; co-organizer, LLMTrust 2026. *Role:* reviewing and PC coordination.

Amit Kumar Sikder (Iowa State) is Assistant Professor of Computer Science working on security and privacy of connected healthcare systems [18] and threat modeling for AI-integrated systems, with publications at IEEE S&P, USENIX Security, ACM CCS, and NDSS; previously Research Scientist, Georgia Tech School of Cybersecurity and Privacy. *Role:* agent-safety and threat-modeling thread.

Shahnewaz Karim Sakib (UT Chattanooga) is Assistant Professor of Computer Science and Engineering working on privacy-preserving and federated learning, fairness, explainability, machine unlearning, and hallucination and adversarial factuality in LLMs [19]. Publications in IEEE TIFS and TDSC; TPC Chair, IEEE DISTILL 2025; leads AI ethics education through Chattanooga’s AI Network. *Role:* privacy, unlearning, and hallucination track lead.

Anindya Bijoy Das (Akron) is Assistant Professor of Electrical and Computer Engineering working on federated learning, LLM trustworthiness in healthcare [20], and distributed computation, with NSF support. PhD, Iowa State (Karas Award); postdoc at Purdue; TPC Chair, DISTILL 2025. *Role:* federated and multi-institution evaluation thread.

Shibbir Ahmed (Texas State) is Assistant Professor of Computer Science working on software engineering for trustworthy AI and reliable deep learning, including LLMs for medical analysis [21], with publications at ICSE, FSE, EMSE, and ICIP and support from NSF and the U.S. Census Bureau. Co-organizer, IEEE DISTILL 2025 and LLMTrust 2026 (with ACM FSE 2026). *Role:* general chair; logistics and proceedings.

Collective experience. The committee has organized two first-edition workshops in two years (DISTILL 2025; LLMTrust 2026), covering reviewer coordination, PC management, OpenReview configuration, and on-time proceedings delivery. All organizers have agreed to the workshop non-compliance policy. No organizer is on a concurrent 2027 workshop proposal.

Program Committee

† has already agreed to serve. The PC will be expanded to 70+ members before the first call for papers.

- Rahul Singh†, Assist. Prof., Univ. of Rhode Island
- Dalal Alharthi†, Assist. Prof., Univ. of Arizona
- Praneeth Narayanamurthy†, Sr. Data Scientist, Alcon
- Niloufar A. Talemi†, Research Assoc., Clemson Univ.
- Sayeed S. Chowdhury†, Assist. Prof., Indiana Univ.
- Ratna Kandala†, Postdoc, Univ. of Kansas
- Mahmoud Nazzal†, Assist. Prof., Old Dominion Univ.
- Farah Ferdaus†, Assist. Prof., Lamar University
- Akram Mohammed†, Bioinfo. Manager, UTHSC
- Aruna Jayasena†, Assist. Prof., UMass Amherst
- Filiz Bunyak Ersoy†, Assist. Prof., Univ. of Missouri
- Zhonghao Liao†, Assist. Prof., Milwaukee Sch. of Engr.
- Silpa Babu†, Postdoc, Univ. of Iowa
- Amit Seal Ami†, Assist. Prof., Univ. of South Florida

- Nigar Khasayeva[†], PhD student, Georgia State Univ.
- Breno Dantas Cruz[†], Postdoc, Tulane University
- An Wang[†], Assist. Prof., Case Western Reserve Univ.
- Erman Ayday[†], Assoc. Prof., Case Western Res. Univ.
- Nasik M. Nafi[†], Postdoc, ORNL
- Emad Shihab[†], Professor, Concordia University
- Sayma Sultana[†], Postdoc, Tulane University
- Gias Uddin[†], Assoc. Prof., York University
- Kevin Xu[†], Assist. Prof., Case Western Res. Univ.
- Jaydeb Sarker[†], Assist. Prof., Univ. Nebraska Omaha
- Abdullah Al Maruf[†], Assist. Prof., Cal. State Univ.
- Yu Yin[†], Assist. Prof., Case Western Reserve Univ.
- Iftexhar Ahmed[†], Assoc. Prof., UC Irvine
- Sanmukh Kuppannagari[†], Assist. Prof., CWRU
- Masud Rahman[†], Assoc. Prof., Dalhousie University
- Soumya Ray[†], Assoc. Prof., Case Western Res. Univ.
- Jing Ma[†], Assist. Prof., Case Western Reserve Univ.
- Yining She[†], PhD student, Carnegie Mellon University
- Shuai Xu[†], Assoc. Prof., Case Western Reserve Univ.
- Feixiong Cheng[†], Postdoc Fellow, Cleveland Clinic
- Chao Yan[†], Assist. Prof., Vanderbilt University
- Yinghui Wu[†], Assoc. Prof., Case Western Reserve Univ.
- Jing Li[†], Professor, Case Western Reserve University
- Md. H. Shahriar[†], Assist. Prof., Univ. of Arkansas

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